

Adam Shaar

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Education

University of Illinois at Chicago <i>Bachelor of Science in Computer Science</i>	May 2024 Chicago, IL
University of Illinois at Chicago <i>Master of Science in Computer Science</i>	December 2025 Chicago, IL

Experience

Junior Solution Architect <i>Andrew Morgan</i>	Oct 2026 – Present <i>Alexandria, VA (Remote)</i>
Designing and building AI solutions on AWS for federal agencies: LLM workflows, retrieval-augmented generation (RAG), and the data pipelines and APIs behind them.	
Software Developer <i>Fisher Printing</i>	Jul 2024 – Oct 2026 <i>Bridgeview, Illinois</i>
- Sole developer of FisherAI, which cut a manual 2-hour shelf-card task to ~5 minutes (24× faster); in daily use at two company branches. - Shipped 190+ releases with signed auto-updates and GitLab CI; automated Adobe InDesign through ExtendScript. - Identified and remediated database security vulnerabilities, hardening protection of sensitive client data.	

Projects

SRMaps: Verified RAG on AWS Bedrock <i>Python, AWS Bedrock, pgvector, PostgreSQL, Terraform</i>	
- Built a retrieval-augmented Q&A and summarization pipeline over 4,300+ geopolitical events: MiniLM embeddings in pgvector, a measured relevance cutoff, and cited answers from gpt-oss-120b. - Added a verification loop in which a second model family (GLM-5) checks every claim against its cited source, with one revision and fail-closed withholding; calibrated the judge against hand labels (10/12 agreement). - Compared 7 prompt and model setups on 76 events: summary faithfulness rose from 33% to 88% (±7 pts) at about \$0.004 per verified answer. - Deployed on AWS (Lambda containers, private RDS Postgres + PostGIS, EventBridge, ECR, GitHub OIDC); wrote the stack as Terraform.	
FisherAI – AI Shelf Card & Excel Generator <i>Rust, TypeScript, Tauri, ONNX Runtime</i>	
- Tauri desktop app that turns sale ads into InDesign shelf cards using YOLO detection, PaddleOCR with dictionary-validated correction, LayoutLMv3, and BiRefNet cut-outs, accelerated with CUDA and Apple CoreML. - Built an authenticated REST inference server (Rust, axum) to offload models from slower machines; validated OCR repair against PDF text layers (3,891 elements).	
Causal Effect of Query Complexity on Product Relevance <i>Python, EconML, scikit-learn, SBERT</i>	<i>Case study</i>
- Built an end-to-end causal inference pipeline on Amazon’s 793K-row multilingual ESCI search dataset (US/ES/JP) to estimate the causal effect of title–query token overlap on exact-match relevance. - Implemented six estimators (adjusted logistic, IPW, AIPW, DRLearner, CausalForestDML) over a domain-informed DAG (PC-Stable), with SBERT embeddings and locale-aware NLP preprocessing. - Measured a robust +14.8–16.3% causal lift (vs. 10.7% naive); text complexity drives 95% of effect variation.	
Reliability Beyond Accuracy: Factuality, Uncertainty & Bias in LLMs <i>Research Paper, UIC</i>	
Authored a review of four LLM-reliability papers (DoLa, KWTDK, BOLD, HELM) on factuality, uncertainty calibration, and bias; proposed benchmarks for jointly measuring hallucination and ideological framing.	

Technical Skills

Languages: Rust, Python, TypeScript, C++, C, C#, F#, Java, SQL, HTML, CSS
Cloud & AI: AWS (Bedrock, Lambda, RDS, ECR, EventBridge, IAM), Terraform, Docker, RAG, LLM evaluation, LangChain, PyTorch, TensorFlow, ONNX Runtime
Technologies: PostgreSQL (pgvector, PostGIS), React.js, Node.js, Express.js, Spring Boot, Tailwind, Firebase, Git, GitHub

Certifications

AWS Cloud Practitioner (CLF-C02) – Exam Prep	Udemy, 2026
Machine Learning A-Z: AI, Python & R + ChatGPT Prize	Udemy, 2025
The Complete Full-Stack Web Development Bootcamp	Udemy, 2024